Notes: Kaustia and Knüpfer (JFE, 2012)

Peer Performance and Stock Market Entry

Contents

1 Big picture

- **Question.** Do the stock market outcomes experienced by local peers affect whether nonparticipants decide to enter the stock market?
- **Setting.** The paper studies Finland from 1995 to 2002 using administrative data on the stock holdings and trades of the full population of direct stock investors. The empirical unit is a zip code by month panel.
- **Core challenge.** A positive relation between local peer returns and local entry could reflect true social influence, but it could also come from market-wide sentiment, local wealth shocks, local firms doing well, media coverage, or household members trading for one another.
- **Main conceptual contribution.** The paper focuses on *outcome-based* social influence rather than the more common action-based peer effect. What matters here is not just whether peers participate, but whether peers have recently done well.
- **Main hypotheses.** Peer outcomes can matter through two broad channels:
  - *naïve extrapolation:* people infer high future returns from peers' recent gains;
  - *selective communication:* people hear more about good than bad experiences, so favorable peer outcomes spread more easily.
- **Main empirical design.** The authors relate next month's zip-code stock market entry rate to last month's zip-code portfolio return, controlling for zip-code fixed effects and province-month fixed effects. The identifying variation is therefore within the same zip code over time, net of province-wide time shocks.
- **Main baseline result.** Higher neighborhood returns predict more stock market entry in the following month. In the baseline specification, a one-standard-deviation increase in neighborhood return raises the entry rate by about 9–13% relative to the mean.
- **Key heterogeneity result.** The effect is much stronger in areas with higher stock market participation, consistent with more opportunities for social learning where more people already invest.
- **Key asymmetry result.** Positive peer returns matter; negative peer returns do not. This asymmetry is central to the interpretation of the paper and is most naturally consistent with selective communication.
- **Main credibility result.** The estimated effect survives a long list of alternative explanations: market-wide shocks, local media coverage, local stocks, household purchases, local wealth effects, and short-sale constraints.
- **Economic takeaway.** Peer success stories appear to help propagate investor sentiment and encourage adoption of stock market participation. The paper therefore provides a microfoundation for how participation booms can develop during periods of high market sentiment.

2 Data and measurement (key objects)

2.1 Why Finland is a useful laboratory

- The paper studies a setting in which direct stock ownership was the dominant household channel for equity participation during the sample period.
- In 2000, households held about 25.8 billion euros in direct stock, compared with only 2.6 billion euros in equity mutual funds and roughly 0.75 billion euros in voluntary pension equity. Directly held stock therefore accounted for about 89% of household stock market investment.
- Direct stock market participation increased from 9.3% in 1995 to 13.9% in 2002, with especially strong growth from 1998 to 2000.
- This makes Finland a good setting for studying first-time stock market entry through direct holdings rather than through mutual funds or pension products.

Interpretation. The empirical outcome—entry into direct stockholding—captures the main participation margin that mattered in Finland during the late-1990s boom.

2.2 Data source and sample construction

- The main data come from the **Finnish Central Securities Depository (FSCD)**, an official ownership registry that records every transaction in the Finnish stock market.
- The sample runs from January 1995 to November 2002.
- The data include investor characteristics, most importantly place of residence, which allows the authors to aggregate holdings and trades to the zip-code level.
- The panel used in the regressions contains **2,648 zip codes** and **93 calendar months**.

2.3 Defining stock market entry

For investor j, the stock market entry date is defined as the first day on which the investor buys stocks of publicly listed companies.

The paper excludes cases that do not reflect an active timing decision by the investor, including:

- equity offerings,
- gifts,
- inheritances,
- divorce settlements,
- and similar transactions with administratively assigned timing.

Interpretation. The dependent variable is meant to isolate *voluntary* entry into the stock market, not passive acquisition of stock positions.

2.4 Neighborhood returns

The key explanatory variable is the return experienced by existing investors in the same zip code. For zip code i and month $t-1$, the baseline neighborhood return is

$$r_{i,t-1} = \frac{1}{N_{i,t-1}} \sum_{j \in i} R_{j,t-1}, \tag{1}$$

that is, the equally weighted average return on the portfolios held by investors residing in zip code i at the beginning of the month.

The paper also considers a value-weighted alternative,

$$r_{i,t-1}^{VW} = \sum_{j \in i} \omega_{j,i,t-1} R_{j,t-1}, \tag{2}$$

where the weights are proportional to portfolio value within the zip code.

Interpretation.

- The equally weighted measure captures how many *people* in the neighborhood had good experiences.
- The value-weighted measure captures the aggregate return on neighborhood wealth.

If the social mechanism works through conversations and social exposure, the equally weighted return should matter more than the value-weighted return.

2.5 Participation rate and other controls

The beginning-of-month stock market participation rate in zip code i is

$$p_{i,t-1} = \frac{\text{number of stock market participants in zip code } i}{\text{number of inhabitants in zip code } i}. \tag{3}$$

This variable is included because:

- areas with many existing investors mechanically have fewer remaining nonparticipants who can enter,
- but at the same time they may generate more opportunities for social interaction and learning.

The authors also merge the securities data with socioeconomic census data from Statistics Finland, including:

- income,
- wealth,
- education,
- population density,
- homeownership,
- self-employment,
- and Swedish-speaking population share.

2.6 Dependent variables

The main monthly entry-rate outcome is measured in basis points and defined in two ways:

$$y_{it}^{pop} = 10{,}000 \times \frac{\text{new investors in zip code } i \text{ in month } t}{\text{total inhabitants in zip code } i}, \tag{4}$$

$$y_{it}^{noninv} = 10{,}000 \times \frac{\text{new investors in zip code } i \text{ in month } t}{\text{inhabitants who are not already investors at the start of month } t}. \tag{5}$$

Interpretation. The first scaling emphasizes population-wide entry; the second isolates entry conditional on being outside the market at the beginning of the month. The two definitions give very similar results.

2.7 Descriptive facts

Table 1 and Table 2 provide the main descriptive facts:

- Average stock market entry over the full 1995–2002 sample is 1.6% at the zip-code level, compared with 1.8% in the aggregate when weighting by population.
- Mean monthly entry rate in the panel is 0.90 basis points.
- Mean monthly neighborhood return is 1.18%.
- Mean participation rate is 9.47%.
- The average cross-sectional standard deviation of neighborhood returns across zip codes within a month is 3.41 percentage points.

Table 1 also shows that stock market entry is systematically higher in zip codes with:

- higher income,
- higher wealth,
- more college education,
- and a larger Swedish-speaking minority population.

Interpretation. The paper is not about who is rich enough to invest; it is about whether *changes* in local peer outcomes shift entry within a zip code over time once these long-run differences are netted out.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper distinguishes between two forms of social influence:

- **Action-based social learning:** people imitate what peers do.

- **Outcome-based social influence:** people react to the outcomes peers have experienced.

The paper's main interest is in the second type. It develops two broad channels:

- **Naïve extrapolation.** Potential entrants update beliefs about expected stock returns from the recent performance of peers, even though peer outcomes are statistically noisy and should be weak signals of fundamentals.
- **Selective communication.** People are more likely to discuss successful investments than unsuccessful ones, so good peer outcomes spread through local interaction more than bad outcomes do.

Interpretation. Both channels predict a positive relation between peer returns and entry, but only selective communication predicts a particularly strong asymmetry between positive and negative returns.

3.2 Baseline panel regression

The main empirical specification is

$$y_{it} = \alpha + \beta r_{i,t-1} + \gamma p_{i,t-1} + u_{it}, \tag{6}$$

where:

- y_{it} is the stock market entry rate in zip code i and month t,
- $r_{i,t-1}$ is last month's neighborhood return,
- $p_{i,t-1}$ is last month's participation rate,
- and u_{it} contains fixed effects and residual shocks.

The baseline estimated equation includes:

- **zip-code fixed effects**, and
- **province-month fixed effects**.

Interpretation.

- Zip-code fixed effects absorb all time-invariant neighborhood characteristics.
- Province-month fixed effects absorb market-wide shocks and province-level shocks in a given month, including media coverage and broad local economic conditions.
- The coefficient β is therefore identified from within-province, within-month differences across zip codes and from within-zip-code changes over time.

3.3 Panel structure and estimation choice

The authors emphasize that the panel is unusually large in both dimensions:

- many cross-sectional units (2,648 zip codes),
- and many time periods (93 months).

They therefore estimate the model with ordinary least squares, following Beck and Katz's arguments for time-series-cross-section settings.

Interpretation. The econometric sophistication in the paper comes less from nonlinear estimators and more from the very rich panel structure plus a careful battery of fixed effects and robustness checks.

3.4 Standard errors

The error terms may be cross-sectionally correlated because social networks and communication can cross zip-code borders.

The paper therefore reports t-statistics under two clustering schemes:

- clustering at the **province-month** level,
- clustering at the **month** level.

The authors also experiment with two-way clustering by month and zip code; it yields standard errors very similar to month clustering.

Interpretation. Month clustering is the more conservative approach because it allows arbitrary cross-sectional correlation across all zip codes within a month.

3.5 Outliers and weighting

Because many zip codes are small, the monthly entry-rate distribution is extremely skewed. The paper therefore checks that the results are not driven by tiny locations with mechanically volatile entry rates.

The three main procedures are:

- trimming observations above the 98th percentile of the entry-rate distribution (the baseline),
- weighting by the inverse of the estimated variance of the entry rate,
- weighting by the number of inhabitants.

Interpretation. The baseline neighborhood-return effect is not an artifact of small places with one or two investors entering in the same month.

3.6 Participation-rate heterogeneity

To test whether social learning is stronger where people have more opportunities to learn from investors, the paper estimates the baseline model separately by quartiles of the lagged participation rate:

$$y_{it} = \alpha_q + \beta_q r_{i,t-1} + \gamma_q p_{i,t-1} + u_{it}, \qquad q \in \{\text{quartiles of } p_{i,t-1}\}. \tag{7}$$

Interpretation. If the neighborhood-return effect is truly social, it should be strongest where more investors are around to share their experiences.

3.7 Positive versus negative peer returns

To distinguish naïve extrapolation from selective communication, the paper estimates a piecewise linear model with a kink at zero:

$$y_{it} = \alpha + \beta_+ \max\{r_{i,t-1}, 0\} + \beta_- \min\{r_{i,t-1}, 0\} + \gamma p_{i,t-1} + u_{it}. \tag{8}$$

A positive-return effect with little or no negative-return effect supports the selective-communication channel.

The paper also estimates a Tobit specification with censoring at zero because the entry rate cannot be negative:

$$y^*_{it} = \alpha + \beta_+ \max\{r_{i,t-1}, 0\} + \beta_- \min\{r_{i,t-1}, 0\} + \gamma p_{i,t-1} + u_{it}, \qquad y_{it} = \max\{0, y^*_{it}\}. \tag{9}$$

Interpretation. The Tobit exercise is not the main design. It is used to show that the asymmetry is not just a mechanical consequence of the dependent variable being bounded below by zero.

3.8 IPO specification

To rule out the idea that new investors simply follow stocks already held locally, the paper re-estimates the neighborhood-return effect using *entry through initial public offerings.*

The regression keeps the same basic logic:

$$y^{IPO}_{i\tau} = \alpha + \beta r_{i,\tau-1} + \gamma p_{i,\tau-1} + u_{i\tau}, \tag{10}$$

but now time is indexed by the start of the IPO subscription period, and the controls include:

- zip-code fixed effects,
- IPO-province fixed effects.

Interpretation. IPO stocks have no local pre-listing return history in neighborhood portfolios, so a positive neighborhood-return effect in IPO entry is hard to reconcile with a "follow the locally held stocks" story.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The paper has no instrument in the usual IV sense. Identification comes from four design features:

- lagged neighborhood returns,
- zip-code fixed effects,
- province-month fixed effects,
- and a panel of all zip codes over 93 months.

Interpretation. The identifying question is: within the same province and month, do zip codes where existing investors just had better returns experience more first-time entry next month?

4.2 Why reverse causality is weak

A natural concern is reverse causality: perhaps stock market entry itself raises neighborhood returns. The paper argues this does not explain the results because:

- the regressions use *lagged* neighborhood returns,
- while price pressure from new entrants would mainly affect contemporaneous returns.

Interpretation. The lag structure rules out the simplest version of "entry causes returns" rather than "returns cause entry."

4.3 Common unobservables and fixed effects

Several potential confounds are eliminated directly by the fixed effects:

- **time-invariant zip-code features** such as financial sophistication, local culture, and baseline investor composition are removed by zip-code fixed effects;
- **market-wide sentiment and province-wide shocks** are removed by province-month fixed effects.

This is especially important for local media coverage because Finnish media markets largely follow provincial borders, so province-month effects absorb newspaper and radio shocks at the relevant level.

4.4 Local stocks and local wealth shocks

The paper takes seriously the concern that neighborhood returns may proxy for local wealth shocks rather than social influence.

The main tests are:

- interacting neighborhood returns with an indicator for municipalities with no local listed stocks,
- interacting returns with an indicator for being more than 30 km from local stocks,
- interacting returns with above-median homeownership,
- and interacting returns with above-median self-employment.

The interaction coefficients are all small and statistically indistinguishable from zero.

Interpretation.

- If local stock performance were driving entry through local wealth, local economy, or local-company attention, the effect should weaken materially in areas without local stocks or should strengthen in places where wealth and income are more locally tied.
- The paper does not find that pattern.

4.5 Household-purchase explanation

Another concern is that one household member may be buying stock for another after personally experiencing good returns. That would not require any social interaction outside the household.

To address this, the paper restricts the sample to males born between 1948 and 1968 and constructs entry and returns within that group, because same-gender same-age pairs of this kind rarely live in the same household.

Interpretation. The neighborhood-return effect survives in this subsample, which makes the "household head buys for spouse or child" story much less convincing as the entire explanation.

4.6 Short-sale constraints and the Tobit check

A subtle alternative explanation is that people hear both good and bad news from peers, but only good news affects entry because negative information is hard to act on when short selling is difficult.

The paper argues against this by noting:

- the unconditional entry rate is positive in the sample, so negative peer outcomes should still reduce entry below that positive baseline if they were discussed symmetrically;
- and the Tobit specification again finds a much larger coefficient on positive returns than on negative returns.

Interpretation. The asymmetry seems to come from how information is transmitted, not just from constraints on how potential entrants can act on bad news.

4.7 What the paper can and cannot distinguish

The evidence strongly supports an effect of peer outcomes, but it does *not* fully separate the relative roles of:

- naïve extrapolation, and
- selective communication with relative-wealth concerns.

The authors remain deliberately agnostic on the precise mix of mechanisms.

Interpretation. The paper is strongest on *whether* peer outcomes matter and on *how the pattern looks*; it is weaker on pinning down the exact psychological mechanism.

4.8 Relation to the literature

The paper's contribution relative to earlier work is threefold:

- It extends stock-market peer-effect research from peers' *actions* to peers' *outcomes.*
- It connects participation booms to local peer success stories rather than to market-wide sentiment alone.
- It provides evidence from a setting where learning from peer outcomes is especially hard to rationalize under strict Bayesian updating, because stock returns are noisy, skill-contaminated, and based on very small samples.

5 Tables and figures

5.1 Figure 1: stock market entry and market returns

Read:

- Figure 1 plots the monthly number of new investors against the cumulative Finnish stock market return from 1995 to 2002.
- The entry series moves strongly with the market boom of the late 1990s.
- At the market peak, entry is roughly five times its average level.

Economic message. This figure motivates the paper. Participation surges in boom periods, and the authors ask whether local peer success stories help explain that pattern.

Table 1
Descriptive statistics.
This table reports descriptive statistics of zip code-level socioeconomic characteristics and stock market entry rates. Panel A summarizes the socioeconomic characteristics of the 2649 zip codes based on census data available at the end of the sample period in 2002. Stock market entry rate, measured in percentage points, is the proportion of inhabitants entering the stock market during the sample period of 1995–2002. Population density is the number of inhabitants divided by the area of a zip code (in kilometers squared). Age measures the average age of all the inhabitants of a zip code. College degrees is the proportion of people with higher academic education. Swedish-speaking measures the proportion of people whose mother tongue is Swedish (Finland has two official languages, Finnish and Swedish). Income and wealth are based on official tax filings of all the individuals living in a zip code. Self-employed is the proportion of inhabitants working as an entrepreneur and homeownership is the proportion of housing units owned by the occupier. Panel B explains stock market entry rates with socioeconomic characteristics. The regressions include decile dummies (except one) for population density, not reported for brevity. Heteroskedasticity robust *t*-values are reported in parentheses below coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: socioeconomic characteristics of zip codes

Characteristic	Mean	Standard deviation	Minimum	25%	Median	75%	Maximum
Stock market entry rate (%)	1.6	1.2	0.0	1.0	1.5	2.0	35.0
Number of inhabitants	1911	2801	101	272	609	2423	24,734
Area (km^2)	110	226	0.1	21	55	114	3511
Population density (per km^2)	299	1010	0.1	4	10	59	23,464
College degrees (%)	55.4	10.0	25.0	48.0	55.0	62.0	92.0
Swedish-speaking (%)	7.0	21.3	0.0	0.0	0.0	1.0	98.0
Income (thousand of euros)	15.6	4.1	9.1	13.0	15.0	17.3	66.4
Wealth (thousand of euros)	52.6	25.2	10.0	39.0	48.0	60.0	397.0
Self-employment (%)	3.4	1.4	0.0	2.5	3.3	4.1	14.6
Homeownership (%)	77.3	15.6	0.0	70.0	82.0	89.0	100.0

Panel B: regressions of stock market entry rate

Variable	(1)	(2)	(3)
Ln(Income)	0.413 (1.85)*	1.107 (4.72) ***	
Ln(Wealth)	0.496 (6.14)***		0.583 (6.78)***
College degrees	0.014 (2.80) ***	0.009 (1.67)*	0.018 (4.38)***
Swedish-speaking	0.014 (7.66) ***	0.014 (7.59)***	0.015 (8.14)***
Number of observations	2,649	2,649	2,649
Adjusted R^2	0.153	0.148	0.151

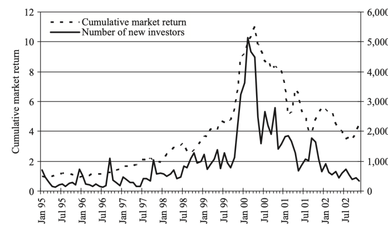

Fig. 1. Stock market entry and market returns. The solid line plots the monthly number of investors who enter the stock market; the dotted line plots the cumulative market return in the Finnish stock market from 1995 to 2002. The stock market entry date is the first day on which an investor buys stocks of publicly listed companies. We exclude entries through equity offerings, gifts, inheritances, divorce settlements, and other transactions that do not represent an active stock market entry decision. The market return is based on the HEX Portfolio Index.

5.2 Table 1 and Figure 2: where entry is high

Read:

- Table 1, Panel A reports cross-sectional zip-code characteristics. The mean zip code has 1,911 inhabitants, wealth of 52.6 thousand euros, income of 15.6 thousand euros, and homeownership of 77.3%.
- Table 1, Panel B shows that richer, wealthier, more educated, and more Swedish-speaking areas have higher stock market entry rates.
- Figure 2 maps stock market entry rates across Finland and shows that entry is concentrated in Southern and Western Finland.

Economic message. There are large, systematic cross-sectional differences in participation. The fixed-effects strategy is crucial because without it these slow-moving regional patterns could be mistaken for peer effects.

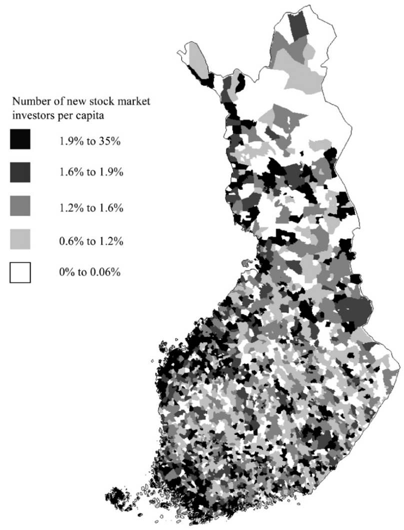

Fig. 2. Distribution of stock market entry rates across zip codes in Finland. This figure plots the stock market entry rates, that is, the number of new investors entering the stock market during the sample period of 1995 to 2002 divided by the total number of inhabitants at the end of 2000, across Finnish zip codes.

Table 2
Past neighborhood returns and stock market entry.

This table shows the results of regressions of the number of investors entering the stock market in a month in a zip code. The dependent variable is the number of new investors entering the stock market divided by either the total number of inhabitants or the total number of inhabitants who do not participate in the stock market, and it is measured in basis points (bps). Neighborhood return is defined as the return on the portfolio of all investors in a zip code, calculated as the average portfolio return of investors living in a zip code. Participation rate is the number of stock market participants divided by the number of inhabitants in a zip code. The regressions include fixed effects for each province-month and each zip code. The regressions are estimated with ordinary least squares, and t-values are robust to clustering at the province-month level (in parentheses) or at the month level (in brackets). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: descriptive statistics

Variable	Mean	Sd	Minimum	25%	Median	75%	Maximum
Entry rate (bps)	0.90	2.72	0.00	0.00	0.00	0.00	32.51
Entry rate for non-investors (bps)	1.05	3.18	0.00	0.00	0.00	0.00	39.53
Neighborhood return (%)	1.18	8.13	−40.05	−3.31	0.94	5.20	99.92
Market return (%)	1.88	10.04	−26.88	−3.66	2.84	8.36	29.37
Participation rate (%)	9.47	6.59	0.00	5.41	7.91	11.24	62.18

Panel B: cross-sectional standard deviation of neighborhood return

	Mean	Standard deviation	Minimum	25%	Median	75%	Maximum
	3.41	1.81	0.99	2.14	2.94	4.00	9.47

Panel C: regressions

	Entry rate (bps)		Entry rate for non-investors (bps)	
Variable	(1)	(2)	(3)	(4)
Neighborhood return	1.056 (4.38)*** [2.03]**	1.042 (4.29)*** [1.98]**	1.190 (4.14)*** [1.99]**	1.190 (4.15)*** [1.98]**
Participation rate		−1.522 (−3.24)*** [−3.14]***		0.005 (0.01) [0.01]
Month-province fixed effects	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Number of zip codes	2,648	2,648	2,648	2,648
Number of observations	246,174	246,174	246,181	246,181
Overall R^2	0.119	0.125	0.121	0.130

5.3 Table 2: the baseline neighborhood-return effect

Read:

- Table 2 is the main baseline table.
- The coefficient on lagged neighborhood return is about 1.04 in the population-scaled entry specification and about 1.19 in the noninvestor-scaled specification.
- The estimate is statistically significant under both province-month and month clustering.
- The implied marginal effect is economically meaningful: a one-standard-deviation increase in neighborhood return raises the entry rate by about 9.4% relative to the mean.

Economic message. Recent local peer gains matter for market entry even after controlling for zip-code fixed effects and province-month shocks.

5.4 Table 3 and Figure 3: social-learning opportunities and asymmetry

Read:

- Table 3, Panel A shows that the neighborhood-return effect rises monotonically across participation-rate quartiles: about 0.64 in the bottom quartile and 1.72 in the top quartile.
- Table 3, Panel B estimates the piecewise-linear model. The coefficient on positive returns is about 1.43–1.65, whereas the coefficient on negative returns is around 0.30 and statistically insignificant.
- Figure 3 plots the category coefficients and shows the same asymmetry graphically; the effect is particularly strong when neighborhood returns exceed 15%.

Economic message.

Table 3
Participation rate interactions and impact of negative and positive returns.

Panel A estimates the model in Table 2 in each participation rate quartile using the total number of inhabitants in scaling the dependent variable (entry rate). The observations are sorted into participation rate quartiles using the beginning-of-month participation rates. Panel B estimates a piecewise linear model that employs a single change in the slope of neighborhood return at zero. The dependent variable is the number of new investors entering the stock market divided by either the total number of inhabitants or the total number of inhabitants who do not participate in the stock market. The regressions include fixed effects for each province-month and each zip code. The regressions are estimated with ordinary least squares, and *t*-values are robust to clustering at the province-month level (in parentheses) or at the month level (in brackets). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Participation rate interactions

	Participation rate			
Variable	Bottom quartile	Second quartile	Third quartile	Top quartile
Neighborhood return	0.641	0.688	1.385	1.717
	(1.74)*	(1.60)	(2.80)***	(3.65)***
	[1.12]	[0.94]	[1.89]*	[1.98]**
Participation rate	−0.419	−6.897	−20.032	−1.538
	(0.24)	(2.40)**	(8.30)***	(2.20)**
	[0.14]	[1.30]	[5.53]***	[2.49]**
Month-province fixed effects	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Number of zip codes	1,190	1,535	1,440	1,050
Number of observations	61,158	61,672	61,694	61,650
Overall R^2	0.089	0.098	0.113	0.135

Panel B: Regressions employing a change in the slope at zero

P	Entry rate (bps)	Entry rate for non-investors (bps)
Variable	(1)	(2)
Max (Neighborhood return, 0)	1.434	1.653
	(4.65)***	(4.55)***
	[2.44]**	[2.50]**
Min (Neighborhood return, 0)	0.302	0.318
	(0.69)	(0.64)
	[0.29]	[0.27]
Participation rate	−1.634	−0.128
	(−3.32)***	(−0.20)
	[−3.28]***	[−0.22]
Province-month fixed effects	Yes	Yes
Zip code fixed effects	Yes	Yes
Number of zip codes	2,648	2,648
Number of observations	246,174	246,181
Overall R^2	0.131	0.138

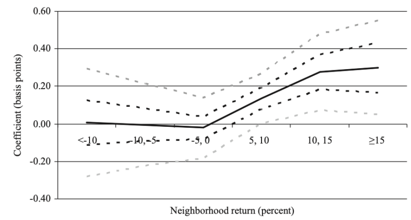

Fig. 3. Influence of neighborhood returns on stock market entry. This figure plots coefficients of dummy variables that each represents a 5% interval of the neighborhood return (0–5% omitted). The dummy variables replace the neighborhood return variable in a regression similar to that in column 2 of Table 2. The black solid line plots the coefficients on the dummy variables, and the dotted lines show the upper and lower bounds of the confidence intervals at the 5% significance level. The black dotted line comes from the standard errors clustered at the province-month level; the gray dotted line from clustering at the month level. The coefficient values indicate, for each return category, the absolute change in the monthly entry rate from the omitted category in basis points.

- The quartile pattern supports the social-learning interpretation.
- The positive-only response supports selective communication.

5.5 Table 4: robustness and alternative explanations

Read:

- Specification 1 replaces equal weighting with value weighting. The coefficient falls to about 0.56, roughly 45% below the baseline.
- Specification 2 uses the same-age same-gender male subsample; the effect survives and implies an increase in entry of about 8% relative to that subsample's mean.
- Specifications 3–6 interact neighborhood returns with no-local-stocks, proximity to local stocks, homeownership, and self-employment; the interaction terms are all small.
- Panel B reports the Tobit model: the positive-return coefficient is about 3.45, the negative-return coefficient is about 0.09.

Economic message.

- Many investors experiencing good outcomes matters more than aggregate neighborhood wealth alone.
- The effect is not driven by local stocks, local wealth, or purchases within households.
- The asymmetry does not disappear when censoring is modeled explicitly.

Table 4
Additional tests and robustness checks.

This table reports additional tests and robustness checks that modify the baseline regression in Table 2. Specification 1 in Panel A weights the individual portfolios within a neighborhood with portfolio value instead of giving them equal weights. Specification 2 estimates the model in a subsample of individuals of the same age and gender, that is, males born between 1948 and 1968. Specifications 3 through 6 interact the neighborhood return variable with an indicator for municipalities with no local stocks, zip codes in which the distance to a local stock is less than 30 km (km), zip codes with above median homeownership rates, and zip codes with above median ratios of self-employed people. The t-values are robust to clustering at the province-month level (in parentheses) or at the month level (in brackets). Panel B reports the results of a Tobit regression that assumes censoring at entry rate of zero. Many province-month blocks have no variation in the entry rate across zip codes, which would require dropping a considerable number of the observations, as required by the Tobit model. Province-month fixed effects are thus replaced with month fixed effects. Zip code fixed effects are included as in the baseline analysis. The model is estimated using unconditional fixed effects for zip codes and months as in Greene (2004). The t-values are robust to clustering at the month level (in brackets). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: interactions and robustness checks

	Neighborhood return		Interaction variable		Participation rate		Number of observations
Specification	Coeff.	t-value	Coeff.	t-value	Coeff.	t-value	
(1) Value-weighted return	0.560	(3.71)*** [2.00]**					246,174
(2) Males born between 1948 and 1968	0.230	(4.07)*** [1.91]*			−3.441	(11.02)*** [9.33]***	241,292
(3) No local stocks	1.042	(2.67)*** [1.12]	0.024	(0.08) [0.02]	−1.522	(3.15)*** [3.27]***	246,174
(4) Distance to local stocks less than 30 km	1.021	(3.28)*** [1.52]	0.040	(0.17) [0.09]	−1.524	(3.15)*** [3.28]***	246,174
(5) Above median homeownership	0.980	(3.62)*** [1.45]	0.190	(0.83) [0.25]	−1.516	(3.13)*** [3.19]***	246,174
(6) Above median self-employment	1.007	(4.17)*** [2.00]**	0.133	(0.94) [0.87]	−1.518	(3.13)*** [3.23]***	246,174

Panel B: Tobit regressions

	Maximum of (Neighborhood return, 0)		Minimum of (Neighborhood return, 0)		Participation rate		Number of observations
Specification	Coeff.	t-value	Coeff.	t-value	Coeff.	t-value	
(7) Tobit assuming censoring at zero	3.452	[1.59]	0.094	[0.02]	−3.616	[2.87]***	234,038

Table 5
Past neighborhood returns and stock market entry through initial public offerings.

This table reports the results of the regressions on the number of investors entering the stock market through an IPO. The dependent variable is the number of new investors entering the stock market through an IPO divided by either the total number of inhabitants or the total number of inhabitants who do not participate in the stock market, and it is measured in basis points (bps). The sample contains 57 IPOs. Independent variables are from the beginning of the subscription period of an IPO. Neighborhood return is the return on the portfolio of all investors in a zip code, calculated as the average portfolio return of investors living in a zip code. Participation rate is the number of stock market participants divided by the number of inhabitants in a zip code. The regressions are estimated with ordinary least squares, and include IPO-province and monthly fixed effects. t-values are robust to clustering at the province-IPO level (in parentheses) or at the IPO level (in brackets). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: descriptive statistics

Variable	Mean	Sd	Minimum	25%	Median	75%	Maximum
Entry rate (bps)	0.61	2.93	0.00	0.00	0.00	0.00	27.01
Entry rate for non-investors (bps)	0.70	3.40	0.00	0.00	0.00	0.00	51.15
Neighborhood return (%)	2.22	8.94	−43.10	−3.09	1.50	8.31	163.45
Participation rate (%)	9.58	6.68	0.25	5.42	7.96	11.38	51.20

Panel B: regressions

	Entry rate (bps)		Entry rate for non-investors (bpp)	
Variable	(1)	(2)	(3)	(4)
Neighborhood return	1.834 (2.57)** [0.70]	1.795 (2.47)** [0.67]	1.852 (2.33)** [0.64]	1.805 (2.22)** [0.61]
Participation rate		0.855 (0.64) [0.47]		1.689 (1.03) [0.75]
IPO-province fixed effects	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Number of zip codes	2,648	2,648	2,648	2,648
Number of observations	148,713	148,713	151,448	151,448
Overall R^2	0.284	0.284	0.279	0.281

5.6 Table 5: IPO entry as a placebo-style test

Read:

- Table 5 looks at entry into the stock market through IPO subscriptions.
- The coefficient on neighborhood return is about 1.80 in the preferred specification.
- The implied effect is roughly a 26.5% increase in IPO-related entry relative to the mean.
- The sample contains 57 IPOs, so month-clustered t-statistics are noisy, but the magnitude is large.

Economic message. The result is difficult to reconcile with a simple "people buy the local stocks that just went up" story, because IPOs have no pre-listing performance in neighborhood portfolios.

6 Short summary

- The paper studies whether peers' *stock market outcomes* affect first-time stock market participation.
- Using Finnish administrative data and a zip-code-by-month panel, it finds that higher recent returns experienced by local investors predict higher local stock market entry next month.
- The estimate is economically meaningful and survives rich fixed effects plus many robustness checks.
- The effect is stronger where local participation is higher, consistent with better opportunities for social learning.
- Only positive peer returns matter, which supports a selective-communication interpretation.
- Local stocks, local wealth, media coverage, household purchases, and short-sale constraints do not explain the main pattern.

- The broader implication is that peer success stories can propagate sentiment and help fuel participation booms.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that recent stock market gains experienced by local peers increase the probability that nonparticipants enter the stock market. This effect is not just a cross-sectional pattern; it appears in panel regressions with zip-code and province-month fixed effects and survives a large set of alternative explanations.

7.2 Economic intuition

People with no personal stock market experience may look to the outcomes of nearby investors as a cognitively accessible source of information. If good experiences are especially likely to be discussed, then local booms in realized returns create local waves of market entry. This mechanism is strongest where more investors are present and where opportunities for social learning are therefore better.

7.3 Takeaway

The paper's message is not merely that peers matter. It is that *peer outcomes* matter, especially good ones, and that these outcomes can shape financial-market participation in a way that helps explain how investor sentiment spreads through populations.